Automatic Street Light Control System

① LDR [Light dependent Resistor]

↳ A passive sensor whoose resistance changes with light intensity.

Working Principal:

Light photons → ↑ charge carriers → ↓ resistance

(Bright light → low R
Darkness → High R)

~~② Voltage Divider~~

LDR is used in Voltage divider.

$$V_{out} = V_{cc} \cdot \frac{R}{R + LDR}$$

Converts lights → analog voltage

② ADC0804 [Analog to digital convertor]

8bit ADC

analog voltage (0-5V) to digital (0-255)

Why Needed? [Microprocessor only works on digital data]

Vin+/Vin- Input

D0-D7 - output data

WR - start conversion

RD - read data

INTR - conversion complete

CLK IN - clock in

Working steps:
WR = LOW (start conversion)
ADC process light
INTR = Low → done
RD = Low → data available on D0-D7

(3) 8085 Microprocessor.
8bit data bus
16 bit address bus → 64KB
Operates on +5V

ALE → seperates address / data
IO/M → memory or I/O
RD / WR → Control signals
INTR → interrupt from ADC

- Reads ADC value
- compares with the threshold
- controls street light

(4.) Latches.

→ AD0 - AD7 multiplexed
→ use Latch to store address
ALE = high [latch stores address]
ALE = low [output remains fixed]

(5.) Memory [EPROM + RAM]

→ Permanent memory
→ Stores program

→ temporary memory
→ stores temporary data.

(6.) ULN2003 (Driver IC)
→ Darlington transistor array [7 channels]

Purpose: Amplifies current

Why Needed:

8085 Output → very low current
Relay needs higher current to ON

Working:
Input high → output sinks current → relay ON.

(7.) Relay
→ Electrically operated switch

Working:

Coil Energized → contacts switch

(8.) 8255 PPI
→ Acts as bridge between 8085 and External world.

Handles:
- Input from ADC0804
- Output to ULN2003

Port A (PA0 - PA7) reads 8-bit data.
Port B (PB0 - PB7) connected to ULN.

Mode-0 (simple I/O mode)
→ No handshake, direct I/P / O/P

Step-by-step working of project

Step-1 Initialization
8085 sends control word → configures 8255

Step-2 Read Light Data
Light photons → Bright / Dark → LR / HR

Data transfer to 8085 ← 8255 reads data from port A ← 8085 triggers read ← INTR LOW ← ADC converts

Step-3 Descision -Making
if value < threshold → Dark
Else Light

Step-4 Output control
8085 writes to PORT B of 8255
E.g PB0 = 1 → Relay ON → Light ON
PB0 = 0 → Relay off → Light OFF

Signal flow :- ADC → Port A (8255) → 8085 → Port B (8255) → ULN → Relay → Light

Assumption:
8255 mapped as I/O Ports
Port A → 00H
Port B → 01H
Port C → 02H
control signal → 03H
control word = 82H

8085 Program:-

```
Initialize 8255   MVI A,82H   control word
                  OUT 03H
START:            IN 00H   (Read ADT)
                  CPI 80H  compare)
                  JC DARK
                  MVI A, 00H  (PB0 =0 , Light off)
                  OUT 01H
                  JMP START
DARK:             MVI A,01H
                  OUT 01H
                  JMP START
                  (PB0 =1) Light = on
```